ANTOCCI ENGINEERING · DWG U1-16 · R9

RAMP SPEC: LOADING DOCK

A 2 kg crate slides on a 30° ramp. g = 9.8 m/s². Weight = 19.6 N.

CALC 1 · DOWN-RAMP PART

$mg \sin \theta = 19.6 \times \sin 30° =$ ____ N

CALC 2 · INTO-RAMP PART

$mg \cos \theta = 19.6 \times \cos 30° =$ ____ N

DRAWING · RAMP FBD: weight straight down, normal ⊥ surface, friction up the ramp. Dashed component guides.

CALC 3 · NO FRICTION

$a = (mg \sin \theta) \div m =$ ____ m/s² down the ramp

CALC 4 · WITH 4 N FRICTION

$F_{net} =$ ____ − 4 = ____ · a = ____

ENGINEER'S NOTE

Is the normal force equal to the weight on this ramp? Explain for the dock crew.

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: net force from the FBD

16 C

DIRECTIONS · Solve each station. Copy the letter into the lock.

STATION 1

Downhill part 6 N, friction 2 N uphill, m = 2 kg. a?

(A) 4 m/s² (T) 2 m/s²

STATION 2

Downhill part 8 N, friction 2 N, m = 3 kg. a?

(I) 2 m/s² (O) 3 m/s²

STATION 3

Down-ramp part of weight is...

(L) mg sin θ (N) mg cos θ

STATION 4

2 kg on 30°: mg sin θ (g = 9.8)?

(T) 9.8 N (P) 17.0 N

STATION 5

Frictionless 30° ramp. a = g sin θ?

(E) 4.9 m/s² (U) 9.8 m/s²

STATION 6

Normal force on a ramp equals...

(D) mg cos θ (S) mg

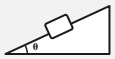
LOCK 1 2 3 4 5 6

Hint: how a ramp's surface sits.

R9 1 2 3

Say it, then write it: forces on a ramp

Language goal: I can **explain** which way it accelerates.



incline
plano inclinado · plano inclinado
a ramp



perpendicular
perpendicular · perpendicular
at a right angle



parallel
paralelo · paralelo
along the same line



net force
fuerza neta · força resultante
all forces added up

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: perpendicular · parallel · down · net · acceleration

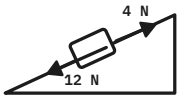
On a ramp, the normal force is ____ to the surface. Part of the weight is ____ to the ramp and points ____ the ramp. Friction points up the ramp. The ____ force decides the direction of the ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The net force points ____, so the box accelerates ____ .
The normal force is not straight up because ____ .

3 · SOLVE WITH THE STEPS



A 2 kg box on a ramp: 12 N down the ramp, 4 N friction up the ramp.

STEP 1 · NET FORCE
____ − ____ = ____ N down

STEP 2 · DIVIDE
a = ____ ÷ ____ = ____ m/s²

STEP 3 · SAY IT
"The box speeds up at ____ m/s², down the ramp."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ The box speeds up because the net force ____ .
☐ I split the weight into parts because ____ .
☐ First I find the ____ force, then the ____ .

I said it ☐ · I wrote it ☐

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every ramp

16 M

DIRECTIONS · Read, draw the FBD, solve along the ramp, then write.

PROBLEM 1 · A 2 kg block has a 7 N downhill weight component and 3 N friction uphill. Find its acceleration.

1 READ

m ____ · downhill ____ · uphill ____

2 DRAW

ramp FBD

3 SOLVE

F_net = 7 − 3 · a = F_net ÷ m

4 WRITE

The block accelerates ____ m/s² ____ because ____ .

PROBLEM 2 · A 2 kg crate is on a frictionless 30° ramp (g = 9.8). Find mg sin θ, then a.

1 READ

m ____ · θ ____ · g ____

2 DRAW

ramp FBD with axes

3 SOLVE

mg sin θ → a = F ÷ m

4 WRITE

The crate speeds up at ____ m/s². The normal force is ____ the weight because ____ .

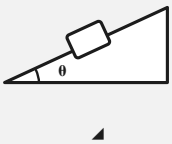
Hardest box: □ Read □ Draw □ Solve □ Write

R9 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: ramps

Language goal: I can **describe** forces on a slope.



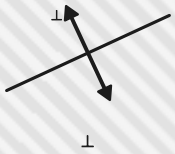
incline

plano inclinado · plano inclinado

a tilted surface

my language: _____

use it: _____



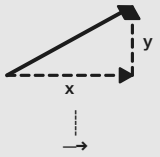
perpendicular

perpendicular · perpendicular

at a right angle

my language: _____

use it: _____



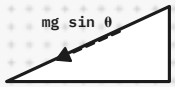
component

componente · componente

one part of a force

my language: _____

use it: _____



mg sin θ

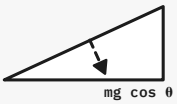
down the ramp

cuesta abajo · rampa abaixo

pulls the block downhill

my language: _____

use it: _____



mg cos θ

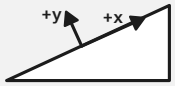
into the ramp

hacia la rampa · contra a rampa

presses into the surface

my language: _____

use it: _____



+x ↘
+y ↗

axis

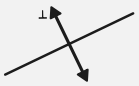
eje · eixo

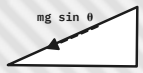
the direction we measure

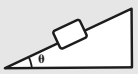
my language: _____

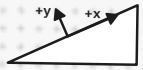
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

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UNIT 1 · DAY 16 · R9 · LOOP CHAIN

I have... who has? Ramps and net force

16 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, and keep going.

★ START — **A**

WHO HAS…

Downhill part of weight on a ramp?

I have **4.9 m/s²** — **B**

WHO HAS…

7 N downhill, 3 N friction, 2 kg: a?

I have **mg cos θ** — **C**

WHO HAS…

2 kg, 30°, g = 9.8: mg sin θ?

I have **m a_x** — **D**

WHO HAS…

Which way does weight always point?

I have **mg sin θ** — **E**

WHO HAS…

Into-ramp part of weight?

I have **straight down** — **F**

WHO HAS…

Does normal force always equal weight?

I have **9.8 N** — **G**

WHO HAS…

That crate, frictionless: a?

I have **2 m/s²** — **H**

WHO HAS…

Each axis gets its own equation: $\Sigma F_x = ?$

I have **no — only if level** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

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